

Investigating the Impact of Alcohol Content on Red Wine Quality

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1. Purpose

1.1 Domain:

Our project focuses on the various chemical factors that contribute to the final perceived quality of red wines.

1.2 Investigation Goal:

Question 1: Is the alcohol content different between high and low quality *vinho verde* red wines?

Question 2: Does the final pH of the wine impact the quality of the *vinho verde* red wine?

1.3 Practical Implications:

The dataset is directly from production plants producing *vinho verde* red wines located in the northwest region of Portugal. Understanding how alcohol content and sugar content impact the perceived quality of red wines will help production plants identify what chemical factors to control for in the production of red wine and help retailers set more accurate prices based on predicted wine quality of the *vinho verde* red wines.

1.4 Population(s) of Interest:

The population of interest is all bottles of *vinho verde* red wines produced by production plants located in the northwestern regions of Portugal.

1.5 Variables of Interest:

Question 1:

- alcohol (%)
 - Percentage alcohol content of the red wine
 - Continuous numeric variable
- quality
 - Perceived quality of the red wine based on sensory tests
 - Ordinal categorical variable from 0 to 10, with 10 being the best
 - Red wines of quality 5 and below are of poor quality
 - Red wines of quality above 5 are of good quality

Question 2:

- pH
 - The pH of the final red wine product
 - Continuous numeric variable

- quality
 - Perceived quality of the red wine based on sensory tests
 - Ordinal categorical variable from 0 to 10, with 10 being the best

1.6 Data Collection Method:

Wine samples (red vinho verde) from the Minho region of Portugal were collected between May 2004 and February 2007 and tested at the official certification entity (CVRVV) using a computerized system (iLab) that recorded both chemical and sensory analyses. Chemical factors such as the fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates and percentage alcohol content are collected during chemical analysis.

The quality of the red wine is evaluated through a standard physicochemical test by at least three sensory assessors, with the final quality scores recorded as the median of the sensory ratings. The quality score is ordinal with 0 being worst and 10 being the best. Each row of the final dataset represents a distinct wine sample with all relevant chemical factors and quality.

2. Data

2.1 Source of Data: [Dataset name, database, website, or organization]

- Dataset name: winequality-red.csv
- Database: Kaggle
- Website: <https://www.kaggle.com/datasets/uciml/red-wine-quality-cortez-et-al-2009>

2.2 Permission/Accessibility

The data is publicly available with no special permission required for usage.

2.3 Time Frame of Data Collection

The *vinho verde* red wine samples were collected over a period of 3 years between May 2004 and February 2007.

3. Topic(s) to Investigate

3.1 Focus of Statistical Investigation:

Question 1: Does high quality *vinho verde* red wines (quality >5) have a significantly higher alcohol content compared to wines with low quality.

Null: The average percent alcohol level of high quality red wines is not significantly different compared to the average percent alcohol level of low quality red wines.

Alternative: The average percent alcohol level of high quality red wines is significantly higher compared to the average percent alcohol level of low quality.

Question 2: Is the pH level of the resultant *vinho verde* red wine significantly correlated with its perceived quality?

Null: The pH level of the final red wine does not have a significant linear relationship with its quality score.

Alternative: The pH level of the final red wine has a significant linear relationship with its quality score.

3.2 Background/Context:

Wine has become increasingly popular beyond luxury markets, and for major exporters like Portugal, ensuring quality is essential for both consumer trust and economic growth. Since taste is difficult to measure and not properly understood, wine quality tests usually combine chemical tests (such as acidity, alcohol, and pH) with expert sensory evaluations. The alcohol content and pH are important parameters that shape the sensory and chemical quality of red wines. Alcohol influences body, warmth, and aroma release, but excessive levels can create imbalance and influence varietal character (Waterhouse et al., 2016). The pH governs acidity perception, color stability, tannin structure, and microbial safety, with lower pH enhancing freshness and long-term stability (Giuffrida et al, 2019). Thus the interplay of alcohol and pH is essential to balance as high alcohol with high pH produces heavy, unstable wines, while balanced values results in vibrancy and harmony (Boulton, 1980; Keller, 2015). Exploring the links between these chemical factors and perceived quality can help winemakers refine their processes and assist retailers in identifying higher-quality wines.

4. Statistical Methods

For question 1, we would use a boxplot to visualize the distribution of percentage alcohol content of the high and low quality wine groups. Using a qqnorm plot we will further confirm that the distribution of percentage alcohol content in both the high and low quality wine groups are approximately normally distributed.

If the sample is normally distributed, we would use the student's t distribution to estimate the 95% confidence interval for the average percent alcohol content in the high and low quality groups. We would then use a Levene's test to see if the percent alcohol content between the two groups are equally distributed. If the equal variance assumption fails we would use the Welch's t-test, if the assumption holds we would use a two sample t-test to see if the average percentage alcohol content in the high quality group is significantly higher than the low quality group.

If the sample is not normally distributed, we would use bootstrapping at 2000 iterations to estimate the 95% confidence interval for the average percent alcohol content in the high and low quality groups. We would then use a bootstrap at 2000 iterations to estimate the 95% confidence interval for the difference in average alcohol content between the high and low quality groups.

For question 2, to visualize our data we would plot the quality of the red wines on the y-axis and the pH of the red wines on the x-axis in a scatterplot. We would then use a residual vs fitted plot and a qqnorm plot of the residuals to test the normality of the residuals and homoscedasticity assumptions. We would then use a linear regression model to see if the quality of the red wine has a significant linear relationship with the pH of the red wine. Finally, we would estimate a correlation value to estimate the strength and direction of this relationship.

References

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